

Turning On the Mixing Board

Module 3 · AR(p), MA(q), and the Fingerprints They Leave

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Previously · the verdict

What Module 2 settled

A whole session on one question — **is this series stationary?** — and no one-number answer, because there isn't one.

- Stationarity is a property of the **process**, not of the one realization you were handed.
- No single exhibit is conclusive. You build a case.
- When the exhibits conflict, the **cost asymmetry** breaks the tie. Spurious regression is the wrongful conviction you can't undo.

Everything today assumes that verdict is already in.



Stationary, precisely

Weak (covariance) stationarity – the three conditions from Module 1:

$$\mathbb{E}[y_t] = \mu, \quad \text{Var} (y_t) = \sigma_y^2 < \infty, \quad \text{Cov} (y_t, y_{t-k}) = \gamma_k$$

The third is important: the covariance depends on the **separation** k , never on the **date** t .

Strict stationarity is the stronger version – the entire joint distribution is shift-invariant:

$$(y_{t_1}, \dots, y_{t_m}) \stackrel{d}{=} (y_{t_1+h}, \dots, y_{t_m+h}) \quad \text{for every } h$$

Strict $\Rightarrow$ weak, given finite second moments. Not the reverse.

We never need it. Every instrument in this course – the correlogram, the DF regression, **both** fingerprints we build today – is assembled out of means, variances, and covariances. **“Stationary” here always means weakly stationary.**

Four exhibits

1. **The plot.** Does it wander off without coming back? Does the spread change? Is there a trend?
2. **The correlogram.** Fast decay $\Rightarrow$ stationary. Slow, near-linear decay $\Rightarrow$ unit root.
3. **The tests, in pairs.** ADF (H_0 : unit root) *and* KPSS (H_0 : stationary) — opposite nulls, so agreement carries information and both failing to reject means the data cannot discriminate.
4. **The ratio.** $\sigma_{\Delta y}/\sigma_y < 0.5 \Rightarrow$ — a sanity check, not a test.

Hold on to exhibit 2. In Module 2 the correlogram was a **yes/no** instrument: does this decay fast enough? Today it becomes an **identification** instrument — and it gets a second barrel.

One case, start to finish



Pick a case, read the four exhibits, then press **Difference it** until the two tests agree. The ladder on the right records what you found at each level.

$I(d)$ is a marker or record, not a definition

$y_t \sim I(d) \iff d$ is the **smallest** number of differences that makes y_t stationary

- $I(0)$ — already stationary. Nothing to do.
- $I(1)$ — one difference does it. Most macro series in levels: GDP, employment, prices.
- $I(2)$ — two. Occasionally price *levels*. Rare, and worth double-checking.

You did not look d up in a table. d is the number of times you had to press the button.

And one press too many is its own diagnosable mistake — that $\rho_1 \approx -0.5$ in firebrick. **Remember what it looked like.** It comes back this later with a name.

Today: the other two letters

$$\text{ARIMA} \left(p, \underbrace{d}_{\text{Module 2}}, q \right)$$

Getting d right is what makes everything downstream “legal”. But d tells you nothing about *what kind* of series you have once you get there.

Difference a random walk and you get white noise — nothing left to model. Difference a real macro series and you get something stationary that still has **structure in it**: this quarter still knows something about last quarter.

That leftover structure is p and q .

Act I · Turning on the board

“Every contact leaves a trace.”

ACT I – MEMORY OF y , MEMORY OF SHOCKS



The board we've been ignoring

Module 1 wrote down the master equation for the whole course:

$$y_t = \alpha + \delta t + \sum_{j=1}^p \phi_j y_{t-j} + \sum_{l=1}^q \theta_l \epsilon_{t-l} + \epsilon_t$$

and then turned almost all of it off. We kept α and ϕ_1 and studied the AR(1) for two modules. The Dickey-Fuller regression is a rearranged AR(1); even the trend specification only added δt .

Term	After Module 2	After today
α	on	on
δt	on (trend ADF)	on
$\phi_1 y_{t-1}$	on	on
$\phi_2 y_{t-2}, ..., \phi_p y_{t-p}$	off	on
$\theta_1 \epsilon_{t-1}, ..., \theta_q \epsilon_{t-q}$	off	on
ϵ_t	on	on

After today the only block you haven't seen is the seasonal one. That's Module 5.

Two kinds of memory

Both halves of the board are memory devices. They remember **different things**.

- **AR memory is memory of y .** *How many days of my own history matter for today?* An $AR(p)$ remembers the last p values of the series; every older shock still reaches the present, but only **indirectly**, through those y 's.
- **MA memory is memory of shocks.** *How many recent surprises are still rattling around in the system?* An $MA(q)$ carries the last q innovations explicitly — and then forgets them.

Your bank balance at the end of day t :

- **$AR(p)$** — today's balance depends on the last p days' balances. Rent, payday, standing orders. The rhythm repeats because your schedule does.
- **$MA(q)$** — today's balance depends on the last q **surprises**: the tax refund, the car repair, the check from your aunt. Once they wash through, they're gone.
- Most real series are **both**.

p counts how far back your own history matters. q counts how long a surprise survives.

AR(p): turning up the rest of the faders

$$y_t = \alpha + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \epsilon_t, \quad \epsilon_t \sim \text{WN}(0, \sigma^2)$$

Collect the lags on the left, exactly as we did for the AR(1) in Module 1:

$$y_t - \phi_1 y_{t-1} - \dots - \phi_p y_{t-p} = \alpha + \epsilon_t$$

$$\underbrace{\left(1 - \phi_1 L - \phi_2 L^2 - \dots - \phi_p L^p\right)}_{\Phi(L)} y_t = \alpha + \epsilon_t \quad \Rightarrow \quad \Phi(L) y_t = \alpha + \epsilon_t$$

The minus signs in $\Phi(L)$ are not a convention chosen to annoy you. **They come for free the moment the $\phi_j y_{t-j}$ terms cross the equals sign.**

Still ϵ_t for the DGP innovation, still e_t for a residual from something we estimated.

When does an $AR(p)$ sit still?

The condition. A stationary $AR(p)$ exists if and only if every root of $\Phi(L) = 0$ lies **outside** the unit circle in the complex plane.

Check it against what you already know. For the $AR(1)$, $\Phi(L) = 1 - \phi_1 L$, one root at $L^* = 1/\phi_1$. “Outside” means $|L^*| > 1$, which is exactly $|\phi_1| < 1$ — the Module 1 condition, unchanged.

The trap. Applying $|\phi_j| < 1$ to the coefficients one at a time is **not** the generalization. Every coefficient can be comfortably less than one and the process still non-stationary — you’ll watch it happen two slides from now.

The condition is on the roots of the joint polynomial. Roots are not coefficients.

AR(2): two roots, one quadratic

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \epsilon_t, \qquad \Phi(L) = 1 - \phi_1 L - \phi_2 L^2$$

Substituting $r = 1/L$ turns the lag polynomial into the **characteristic equation**, where the arithmetic is cleaner and the condition turns inside out:

$$r^2 - \phi_1 r - \phi_2 = 0 \qquad \Rightarrow \qquad \lambda_{1,2} = \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{2}$$

Stationary $\Leftrightarrow |\lambda_1| < 1$ and $|\lambda_2| < 1$. Same statement as “roots of $\Phi(L)$ outside the circle” – the two sets are reciprocals, so the picture just flips. Both conventions are in print; the widget below draws them both.

The discriminant is a one-glance classifier.

$\phi_1^2 + 4\phi_2$	Roots	What the ACF does
> 0	a real pair	decays smoothly – a mixture of two geometrics
< 0	complex conjugates	a damped sine wave – crosses zero, repeatedly

Stationarity is a *separate* question, settled by the modulus, not by the sign of the discriminant.



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The AR fingerprint: it tails off

We won't derive the ACF of a general $AR(p)$ — that's Yule–Walker, and it's in Enders ch. 2. The **pattern** is what we need.

- **AR(1)**: $\rho_k = \phi^k$. Pure geometric decay. You've known this since Module 1.
- **AR(2), real roots**: a mixture of two geometrics — still smooth.
- **AR(2), complex roots**: a damped sinusoid — it crosses zero and comes back.
- **AR(p)**: one geometric or damped-sine term per root, all added together.

The decay **rate** belongs to the largest-modulus root. The **shape** belongs to whether the roots are real or complex.

The ACF of an AR process tails off. It never cuts off.

Half the identification rules in Act III are that one sentence. [Here comes the other half.](#)

The other kind of memory

A hurricane closes a refinery on the Gulf coast. Within a week gas prices are up forty cents everywhere east of Texas. Six weeks later the refinery is back online, and so is the price.

That shock didn't decay. It **expired**. It had a length — however long it takes to bring capacity back — and then it was over.

Ask an AR(1) the same question. Does a shock ever actually leave?

$$y_t = \phi y_{t-1} + \epsilon_t \quad \Rightarrow \quad \frac{\partial y_{t+h}}{\partial \epsilon_t} = \phi^h$$

ϕ^h gets small. It gets unmeasurably small. **It is never zero.**

AR forgets asymptotically. MA forgets absolutely.

Same shock, two systems



Identical innovations top and bottom, one hurricane at t_0 . The ledger beside each series breaks today's value into its pieces; the **firebrick** segment is the part still owed to the shock. **Before we step it: at which period does the hurricane leave each system?**

MA(q) has a **horizon** – at $t_0 + q + 1$ the firebrick bar slides off the register and the series has no memory of the hurricane at all. AR(1) has a **half-life** – the bar shrinks by a factor of ϕ every period and never reaches the edge.

MA(q): the DGP

$$y_t = \alpha + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q} = \alpha + \epsilon_t + \sum_{l=1}^q \theta_l \epsilon_{t-l}$$

In lag-operator form, with $\theta(L) = 1 + \theta_1 L + \theta_2 L^2 + \dots + \theta_q L^q$:

$$y_t = \alpha + \theta(L) \epsilon_t$$

Look hard at the right-hand side and notice what **isn't** there.

There is no y on the right. An MA(q) is built out of nothing but the last $q + 1$ shocks.

Every strange and useful property on the next four slides falls out of that one structural fact. The series is a weighted sum of a **finite window of noise** — so it cannot wander off, and it cannot remember anything older than the window.

Why $\Theta (L)$ has plus signs

Polynomial	Course convention	Where the signs came from
$\Phi (L) - \text{AR}$	$1 - \phi_1 L - \phi_2 L^2 - \dots$	the $\phi_j y_{t-j}$ crossed the equals sign
$\Theta (L) - \text{MA}$	$1 + \theta_1 L + \theta_2 L^2 + \dots$	the $\theta_l \epsilon_{t-l}$ never moved

Enders and Hamilton agree with us. Box, Jenkins and Reinsel write the MA polynomial with minus signs, so their θ is our $-\theta$. **Check the convention before you copy a formula out of a book.** [notation_dictionary.md](#) is the final word for this course.

Put both polynomials on one line and the whole master equation collapses to:

$$\Phi (L) y_t = \alpha + \Theta (L) \epsilon_t$$

That is ARMA(p, q) – and it is the object Module 4 estimates.

MA(q) is always stationary

No conditions on the θ 's. None. Check the three requirements directly:

$$\mathbb{E}\left[y_t\right] = \alpha,$$

$$\text{Var}\left(y_t\right) = \sigma^2(1 + \theta_1^2 + \dots + \theta_q^2),$$

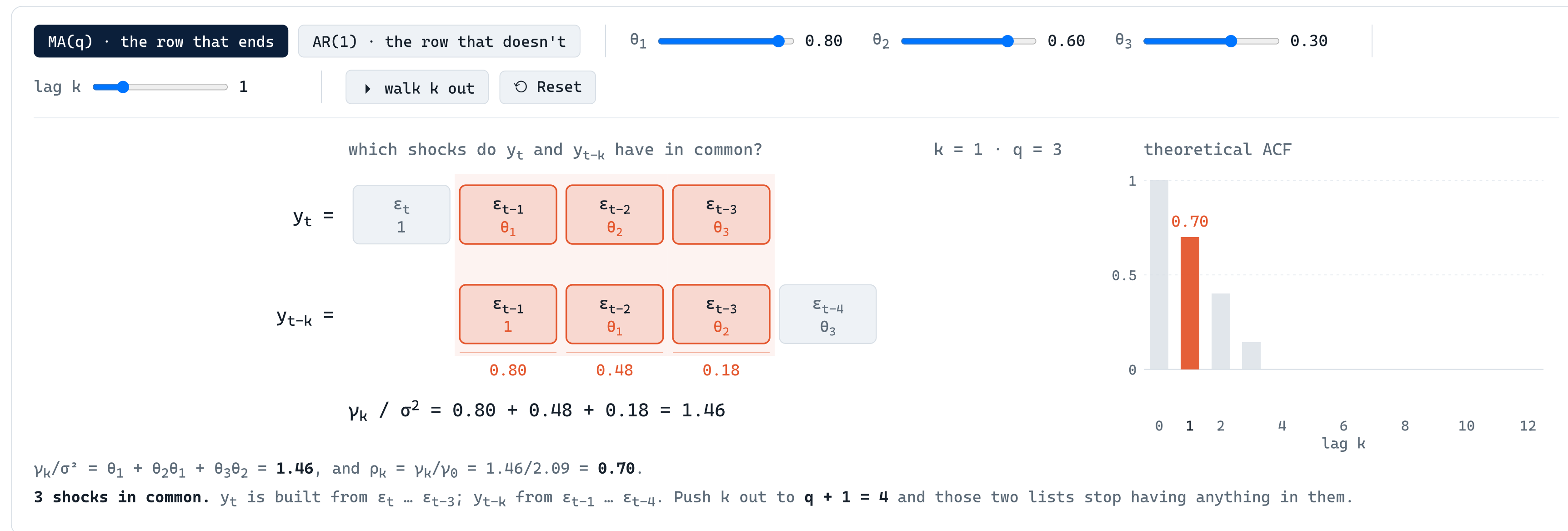
$$\gamma_k = \sigma^2 \sum_{l=0}^{q-k} \theta_l \theta_{l+k} \quad (\theta_0 \equiv 1)$$

The mean is α because every innovation has mean zero. The variance moves with the θ 's but not with t . And γ_k counts how many ϵ 's y_t and y_{t-k} **share** — which depends on the separation k and never on the date.

	AR(p)	MA(q)
Stationarity	a condition — roots of $\Phi(L)$ outside the circle. You can fail it.	automatic , free, no strings
The condition it carries instead	—	invertibility

Stationarity and invertibility are not the same property. They get conflated every single year. Three slides from now you'll see why they can't be the same thing.

Where the cutoff comes from



Two copies of the series, shifted by k . Shared columns light up, the products print underneath, and they sum to γ_k / σ^2 . Walk k up one step at a time and watch the shared columns run out: at $k = q + 1$ the two rows have **nothing in common** and the sum is empty.

Then flip to $\Phi(L)$ — the same diagram built from ψ -weights that never end, so the rows never come apart. **One picture has an edge. The other doesn't. That's the whole difference.**

The algebra, once, by hand

$$y_t = \epsilon_t + \theta \epsilon_{t-1}, \quad \mathbb{E}[\epsilon_s \epsilon_t] = 0 \ (s \neq t), \quad \mathbb{E}[\epsilon_t^2] = \sigma^2$$

$$\gamma_0 = \text{Var} (\epsilon_t + \theta \epsilon_{t-1}) = \sigma^2 + \theta^2 \sigma^2 = \sigma^2 (1 + \theta^2)$$

$$\gamma_1 = \mathbb{E}[(\epsilon_t + \theta \epsilon_{t-1})(\epsilon_{t-1} + \theta \epsilon_{t-2})] = \underbrace{\mathbb{E}[\epsilon_t \epsilon_{t-1}]}_0 + \underbrace{\theta \mathbb{E}[\epsilon_t \epsilon_{t-2}]}_0 + \underbrace{\theta \mathbb{E}[\epsilon_{t-1}^2]}_{\sigma^2} + \underbrace{\theta^2 \mathbb{E}[\epsilon_{t-1} \epsilon_{t-2}]}_0 = \theta \sigma^2$$

$$\gamma_2 = \mathbb{E}[(\epsilon_t + \theta \epsilon_{t-1})(\epsilon_{t-2} + \theta \epsilon_{t-3})] = 0$$

Every term is a product of two **different** innovations. Nothing is shared, so there is nothing to average. The same argument kills γ_k for every $k \geq 2$.

$$\rho_1 = \frac{\theta}{1 + \theta^2}, \quad \rho_k = 0 \text{ for all } k \geq 2$$

A free diagnostic — and a problem

Treat $\rho_1 = \theta / (1 + \theta^2)$ as a function of θ and find its extremes:

$$\max_{\theta} \frac{\theta}{1 + \theta^2} = \frac{1}{2} \text{ at } \theta = 1, \quad \min_{\theta} \frac{\theta}{1 + \theta^2} = -\frac{1}{2} \text{ at } \theta = -1 \quad \Rightarrow \quad |\rho_1| \leq 0.5$$

For **every** MA(1), whatever θ is. So if the sample $|\hat{\rho}_1| > 0.5$, the series cannot be a pure MA(1). No test, no critical value, no software — just algebra you did on the last slide.

Now the problem. Feed θ and $1/\theta$ into the same formula:

$$\rho_1(\theta) = \frac{\theta}{1 + \theta^2}, \quad \rho_1(1/\theta) = \frac{1/\theta}{1 + 1/\theta^2} = \frac{\theta}{\theta^2 + 1}$$

Identical. $\theta = 0.7$ and $\theta \approx 1.43$ produce the same autocorrelation at every lag, so they produce the same correlogram, the same second moments, the same everything we can measure. **The data cannot tell them apart.**

Invertibility: which θ do you get to keep?

Two parameter values, one correlogram. Something has to break the tie, and it isn't the data.

The question invertibility asks. Given the observed history $y_t, y_{t-1}, \dots$, can we recover the innovations ϵ_t that produced it?

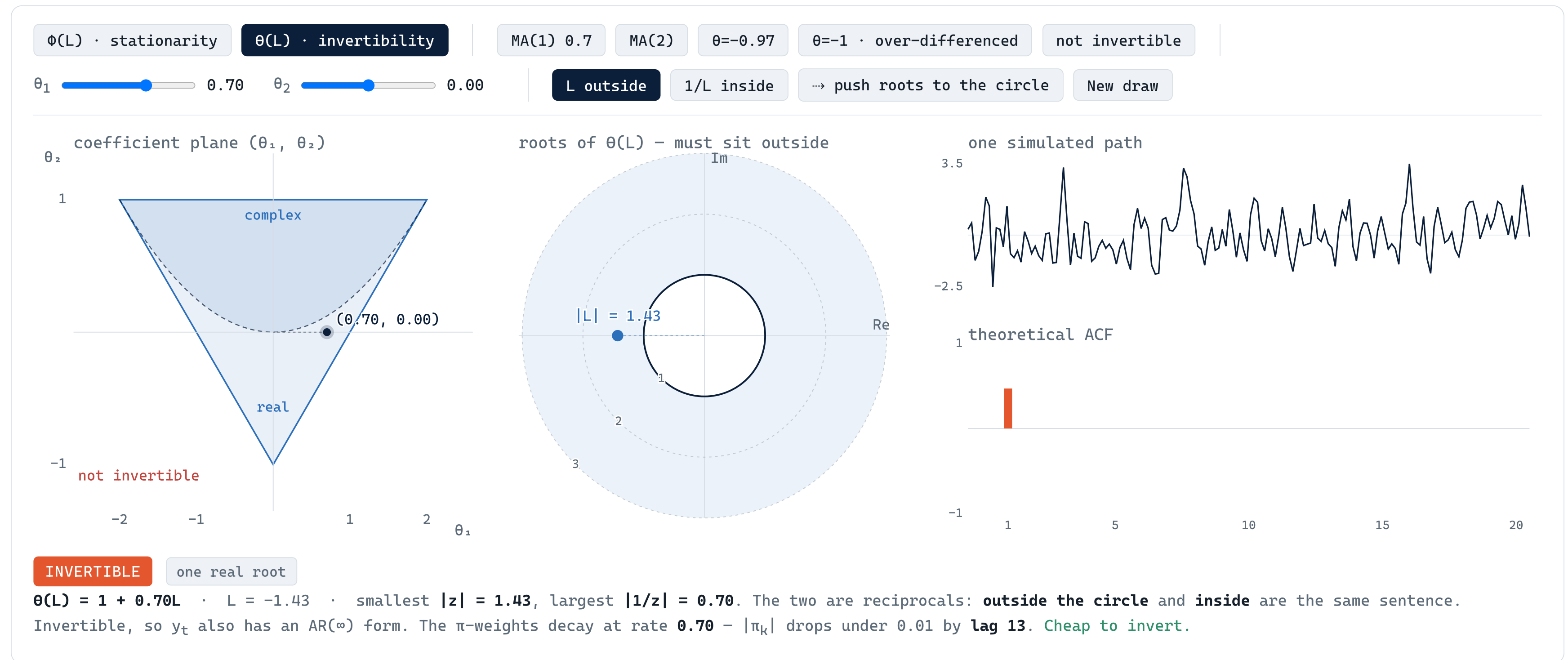
- **Yes** — the representation is unique, the θ 's are identified, and MLE has one peak to climb.
- **No** — two configurations carry the same likelihood, and MLE either refuses to converge or pins the estimate to the boundary.

The condition. An $MA(q)$ is invertible $\Leftrightarrow$ every root of $\theta(L) = 0$ lies **outside** the unit circle.

Side	Property	Polynomial	Roots
AR	stationarity	$\Phi(L)$	outside the circle
MA	invertibility	$\theta(L)$	outside the circle

Same language, same geometry, different polynomial — and **different meaning**. The MA is stationary either way; invertibility is about whether you can *identify* it. **Of the reciprocal pair, exactly one is invertible. That's the one you're allowed to report.**

The mirror, on the board



Same widget, same algebra, one sign flipped: $\Phi(L)$ mode asks *is it stationary*, $\theta(L)$ mode asks *is it invertible*, and the region turns upside down.

At $\theta_1 = 0.70$ the readout says $L = -1.43$: the root sits outside, **invertible**. Note the ACF – one spike at lag 1, nothing after. Now drag the θ_1 **slider** to 1.43: the reciprocal $1/0.7$, the partner from two slides ago.

The ACF panel doesn't move. But the root becomes $L = -0.70$ and crosses inside: **not invertible**. The coefficient and the root have **traded places** – that is what “reciprocal pair” looks like, and it is the tie being broken in front of you.

The over-differencing trap, finally named

Module 2, differencing a series that only had a trend. Start from $y_t = \alpha + \delta t + \epsilon_t$:

$$\Delta y_t = \delta + \epsilon_t - \epsilon_{t-1}$$

An MA(1) with $\theta = -1$. Its polynomial is $\theta(L) = 1 - L$, and the root of $1 - L = 0$ is

$$L = 1 \quad \text{— sitting **on** the unit circle.}$$

Non-invertible. Not “nearly” — on the boundary, by exactly zero margin.

And look where $\theta = -1$ landed two slides ago. It is the precise value that attains $\rho_1 = -0.5$: the floor of the bound.

The firebrick spike you were told to remember is the bound being hit — which is also the edge of invertibility.

So $\hat{\theta}$ pinned at -1 after differencing — with a standard error too good to be true — is the warning light. Two modules ago you took that on faith.

Two ways to write the same dynamics

A finite stationary AR(p) and a finite invertible MA(q) are two spellings of the same kind of memory. Each is the infinite-order version of the other.

One direction you already did. Module 1, recursive substitution on the AR(1):

$$y_t = \frac{\alpha}{1 - \phi_1} + \sum_{i=0}^{\infty} \phi_1^i \epsilon_{t-i}$$

An AR(1) is an MA(∞). **Wold's theorem** says every covariance-stationary process has such a representation.

The other direction is what invertibility buys. Take $\theta = 0.7$, so $\Theta(L) = 1 + 0.7L$:

$$\Theta(L)^{-1} = \frac{1}{1 + 0.7L} = 1 - 0.7L + 0.49L^2 - 0.343L^3 + \dots$$

$$y_t = 0.7 y_{t-1} - 0.49 y_{t-2} + 0.343 y_{t-3} - 0.2401 y_{t-4} + \dots + \epsilon_t$$

$\pi_k = (-\theta)^k$, so $\pi_1 = -0.7, \pi_2 = +0.49$ — alternating signs, decaying at rate $|\theta|$. (The equation above is the rearrangement onto the y side, so its coefficients are $-\pi_k$.) That series converges **because** the root is outside the circle. **That is what invertibility does, operationally.**

Check it in R

An MA(1) has one parameter. Fit it a high-order AR anyway and see what comes back.

```
1 set.seed(8675309)
2 y <- arima.sim(n = 5000, list(ma = 0.7))
3 ar(y, order.max = 8, aic = FALSE, method = "ols")$ar
```

lag k	1	2	3	4	5
fitted $-\widehat{\pi}_k$ (what <code>ar()</code> returns)	0.680	-0.481	0.340	-0.224	0.178
theory $-\pi_k = (-1)^{k+1} (0.7)^k$	0.700	-0.490	0.343	-0.240	0.168

It won't be exact — 5000 draws of noise, and the far tail generally sags below theory (lag 5 here runs a little hot) because a fit truncated at 8 lags has to absorb everything past lag 8. But the **sign alternation** and the **decay at 0.7** are unmistakable, and nobody put them there by hand.

One parameter — $\theta = 0.7$ — generated that entire row.

Small p , small q

Why this matters past the algebra:

- A **small AR** captures dynamics that would need infinitely many MA weights.
- A **small invertible MA** captures dynamics that would need infinitely many AR weights.
- An ARMA(1,1) — two dynamic parameters — describes behavior a pure AR would need six or eight lags to imitate.

A finite ARMA is a compact code for an infinite-order process.

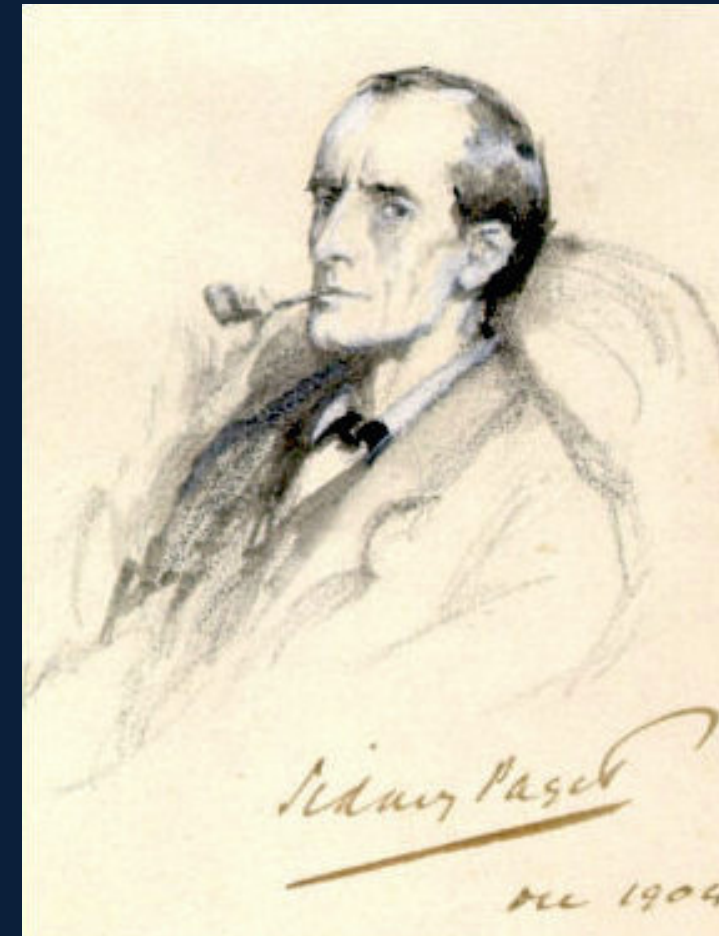
Three places we cash this in:

1. **Module 4 model selection** is largely the hunt for the most parsimonious of several equivalent representations. AR(5) or ARMA(1,1)? Take the ARMA(1,1) — fewer parameters, less overfitting.
2. **Forecasting** from an MA runs on the AR(∞) form underneath, because forecasts use past *observations*, not unobserved shocks. Invertibility is what licenses that substitution.
3. **The ARMA trap in Act III**: both representations are effectively infinite-order, so *neither* correlogram cuts off cleanly. You'll see it, and it's why Module 4 needs information criteria.

Act II · The second fingerprint

“Two prints, or it isn’t an
identification.”

ACT II – THE PACF



One fingerprint isn't enough

Run the ACF across the whole family and look at what comes back:

Process	ACF	Can you read the order off it?
AR(1)	tails off	no
AR(2)	tails off	no
AR(5)	tails off	no
MA(1)	cuts off at 1	yes
MA(2)	cuts off at 2	yes

The ACF is a *complete* instrument for MA order and a *useless* one for AR order. It will tell you “this is an AR” and then refuse to say which one.

Look at the shape of that problem. We don't need a better ACF — we need an instrument that does for AR what the ACF already does for MA: **cut off at the order**.

That instrument exists, and you already own every piece of it. It's OLS.

The partial autocorrelation function

The partial autocorrelation at lag k , written ϕ_{kk} , is the correlation between y_t and y_{t-k} **after removing the linear effect of everything in between** — $y_{t-1}, y_{t-2}, \dots, y_{t-k+1}$.

How much does y_{t-k} tell me about y_t that the lags in between haven't already told me?

The ACF never asks that. ρ_k is the **raw** correlation between y_t and y_{t-k} , and it happily counts influence that arrived *indirectly*, passed down through the lags in between.

That is exactly why an AR(1) has a non-zero $\rho_2 = \phi^2$. There is no y_{t-2} term anywhere in the AR(1) equation — but y_{t-2} moved y_{t-1} , and y_{t-1} moved y_t . The correlation is real, and it is entirely **second-hand**.

The PACF refuses to count second-hand information. That refusal is the whole idea.

The partial-effect interview

You’re predicting whether someone is **late to work today**.

“**Were they late yesterday?**” Useful — lateness is persistent. That’s the lag-1 correlation.

“**Were they late two days ago?**” This also looks useful. But yesterday and two-days-ago are themselves correlated: someone late yesterday was probably late the day before as well. So the honest question is

Does two-days-ago tell me anything **beyond** what yesterday already told me?

The answer to that question is ϕ_{22} .

Process	Does y_{t-2} add anything past y_{t-1} ?	ϕ_{22}
AR(1)	No — y_{t-2} only ever reached y_t through y_{t-1}	0
AR(2)	Yes — y_{t-2} sits in the DGP with its own coefficient ϕ_2	$\neq 0$

Push it out to p : an $AR(p)$ has non-zero partial autocorrelation at lags 1, ..., p and **exactly zero from $p + 1$ onward**. **There is the cutoff we went looking for.**

It really is just OLS

“Remove the linear effect of the intermediate lags” is a phrase you already know how to execute. **Regress on them.** For lag k , run

$$y_t = \hat{\beta}_0 + \hat{\beta}_1 y_{t-1} + \hat{\beta}_2 y_{t-2} + \dots + \hat{\beta}_k y_{t-k} + e_t$$

and keep the coefficient on the **deepest** lag:

$$\hat{\phi}_{kk} = \hat{\beta}_k$$

The sample PACF out to lag K is K regressions, each one lag deeper than the last, reading the bottom coefficient off each.

Why software doesn't do it that way. K separate regressions is wasteful, so `pacf()` pushes the **Yule-Walker** equations through the **Durbin-Levinson** recursion and gets the same numbers far faster. We won't need that machinery — Enders ch. 2 p. 65 has it if you want it.

Keep the regression definition in your head anyway. It's the one that explains what the number *means*.

Build it by hand

```
1 set.seed(8675309)
2 y <- arima.sim(n = 500, list(ar = c(0.6, -0.3))) # an AR(2)
3
4 y_t <- y[5:500]; y_lag1 <- y[4:499]; y_lag2 <- y[3:498]
5 y_lag3 <- y[2:497]; y_lag4 <- y[1:496]
6
7 phi_11 <- coef(lm(y_t ~ y_lag1))[2]
8 phi_22 <- coef(lm(y_t ~ y_lag1 + y_lag2))[3]
9 phi_33 <- coef(lm(y_t ~ y_lag1 + y_lag2 + y_lag3))[4]
10 phi_44 <- coef(lm(y_t ~ y_lag1 + y_lag2 + y_lag3 + y_lag4))[5]
11
12 c(phi_11, phi_22, phi_33, phi_44)
13 pacf(y, lag.max = 4, plot = FALSE) # should match
```

	ϕ_{11}	ϕ_{22}	ϕ_{33}	ϕ_{44}
four <code>lm()</code> calls	0.450	−0.259	0.038	−0.051
<code>pacf()</code>	0.446	−0.257	0.040	−0.052

Agreement to the third decimal; the small gap is `pacf()` lining up the start of the sample a little differently. And lags 3 and 4 sit **inside** the $\pm 1.96/\sqrt{500} = \pm 0.088$ band. **The cutoff isn't a metaphor — it's two numbers that stopped mattering.**

What those numbers actually are

The DGP was $\phi_1 = 0.6$, $\phi_2 = -0.3$. Look at what came back.

$$\hat{\phi}_{22} = -0.259 \quad \leftrightarrow \quad \phi_2 = -0.3$$

The PACF at the final lag estimates the AR coefficient at that lag. Not a coincidence: at $k = p$ the regression you are running is the $AR(p)$ model, so its deepest coefficient is ϕ_p .

Now the one that catches people every year:

$$\hat{\phi}_{11} = 0.450 \quad \text{but} \quad \phi_1 = 0.6 \quad (!)$$

ϕ_{11} is the lag-1 coefficient with **nothing** controlled for, so it is just ρ_1 – and for an $AR(2)$,

$$\rho_1 = \frac{\phi_1}{1 - \phi_2} = \frac{0.6}{1.3} = 0.462$$

– and the sample came back at 0.446. $\hat{\phi}_{11}$ is $\hat{\rho}_1$ exactly, by construction; the 0.462 is the population value it estimates. $\phi_{11} = \phi_1$ **only when the process really is an $AR(1)$** . The PACF hands you the model's *deepest* coefficient, not its first.

Both fingerprints, side by side



AR(1): the ACF tails away, the PACF has **one** spike. **AR(2) real**: the ACF still tails, the PACF has **two**. The PACF is counting p for you. Now press **MA(1)** and watch the two panels **trade jobs**.

Why the MA side tails off

The PACF of an MA(q) does **not** cut off. It decays — geometrically, sometimes oscillating.

You already know why. We inverted an MA(1) a few slides ago:

$$y_t = 0.7 y_{t-1} - 0.49 y_{t-2} + 0.343 y_{t-3} - 0.2401 y_{t-4} + \dots + \epsilon_t$$

An invertible MA(q) **is** an AR(∞). The PACF asks “does lag k add anything new?”, and the AR(∞) representation keeps answering *yes, a little* — at every lag, forever, the amount shrinking geometrically.

There is no finite k at which the answer becomes exactly *no*, because there is no finite k at which π_k becomes exactly zero.

The duality isn’t a curiosity. It’s the reason the second fingerprint works at all.

The mirror

	ACF	PACF
AR (p)	tails off	cuts off at p
MA (q)	cuts off at q	tails off

Read it in either direction and it's the same sentence. An AR is finite in its own lags and infinite in shocks, so it cuts in the PACF and tails in the ACF. An MA is finite in shocks and infinite in its own lags, so it does exactly the reverse.

Two instruments, and between them they pin down **both** orders — as long as the process is pure.

Which raises the obvious question. What does an ARMA(p, q) — finite in neither direction — do to this table?

That's Act III.

Act III · The lineup

“No print is readable until the surface holds still.”

ACT III – IDENTIFICATION

Before you read a print

You asked what an ARMA does to the table. Before we answer that — a prerequisite.

Every fingerprint in this module was computed on a **stationary** series. That was not a convenience. γ_k is only well defined if the covariance depends on the gap k and nothing else, so on a non-stationary series the ACF is not a memory measurement at all.

The order of operations, for the rest of the course:

1. **Verdict first.** Plot it, correlogram it, ADF and KPSS it — Module 2's whole apparatus.
2. **Difference until stationary.** That count is d . Not one more — Act I showed you what over-differencing costs ($\hat{\theta} \rightarrow -1$).
3. **Only now read the ACF and the PACF**, on the stationary series.
4. **Where the two prints disagree or both smear**, Module 4 arbitrates.

Identify d first, then p and q . Never the other way round.

Step 1, on the board



Before you press anything: barely any decay. Enormous memory — or a series that isn't ready to be read? Press **Difference it** once and watch the verdict flip.

A near-1 ACF at every lag is **a question you haven't answered yet**, not a long-memory AR.

The lineup

Three suspects. Two instruments. One table for the rest of the course.

Process	ACF	PACF
$AR(p)$	tails off	cuts off at lag p
$MA(q)$	cuts off at lag q	tails off
$ARMA(p, q)$	tails off	tails off

The pure cases each announce themselves with a sharp cutoff in exactly one panel. That cutoff *is* the order.

The third row is the honest one. An ARMA is finite in neither direction, so **both** prints smear and neither reads out a number. You can tell that it *is* an ARMA — that is genuine information — but you cannot read p and q off the picture.

Row three is why Module 4 exists.

The gallery



Same T , same seed, six processes. **Call the row before I press the button.**

Watch **AR(2) cplx** especially — the ACF *oscillates* on the way down. Still tailing off, still two PACF spikes. **Decaying does not mean monotone.**

Where the cutoff stops being sharp

The table is a statement about the **population**. You never have the population.

Go back to the gallery and drag T from 500 down toward 60. Two things happen at once:

- The Bartlett bands **widen** as $\pm 1.96/\sqrt{T}$ — at $T = 60$ that is ± 0.25 .
- The sample autocorrelations get noisier, so lags past the true cutoff start poking out.

A “cutoff” is a spike outside the band followed by spikes inside it. Widen the band and blur the spikes and an MA(2) can read as an MA(1), or as an ARMA, or as nothing at all.

The fingerprint is evidence, not a confession.

Pushing it until it breaks

Everything so far shows what the theory **promises**. Now push it to the edge and watch where it gives.

Take an MA(1) at $\theta = -0.97$. On paper this is still entirely legal:

$$y_t = \epsilon_t - 0.97 \epsilon_{t-1}, \quad \theta(L) = 1 - 0.97L, \quad L^* = \frac{1}{0.97} = 1.031.$$

The root sits **outside** the unit circle, so the process is invertible and every result from Act I applies. But 1.031 is *barely* outside – and $L^* = 1$ is the over-differencing root.

Meanwhile $\rho_1 = \theta / (1 + \theta^2) = -0.4998$, sitting right on the $|\rho_1| \leq 0.5$ ceiling. **Legal, and one nudge from not being.**

Still legal, on the board



The θ_1 slider stops at -0.95 ; this instance starts past the end of it on purpose.

Read the ACF. **That is a clean MA(1) cutoff** — one big negative spike, then nothing.

The correlogram doesn't warn you

And it reads fine on the numbers too: $\hat{\rho}_1 = -0.475$, lags 2–6 all **inside** the Bartlett band. Identification succeeds. Then you try to *estimate* it:

```
1 set.seed(1999)
2 y_bad <- arma_simulator(n = 500, theta = -0.97)
3 arima(y_bad, order = c(0, 0, 1))
```

	$\hat{\theta}$	s.e.	true θ
MA(1), $\theta = 0.7$	0.732	0.031	recovered
MA(1), $\theta = -0.97$	-1.000	0.009	pegged

MLE slid $\hat{\theta}$ onto the invertibility boundary and stopped — and then reported a standard error of 0.009, which is **confidence, not accuracy**. $\hat{\theta} = -1$ is the over-differencing signature from Act I, arriving here from a completely different direction.

The ACF said MA(1) and was right. The estimator failed anyway.

Blind draw



Press **Blind draw**. Commit out loud before anyone reveals: *which* panel cuts off — ACF, PACF, **neither** — and at what lag?

That answer is the row, and the lag is the order. Again until the room is faster than I am.

Act IV · Back to the case file

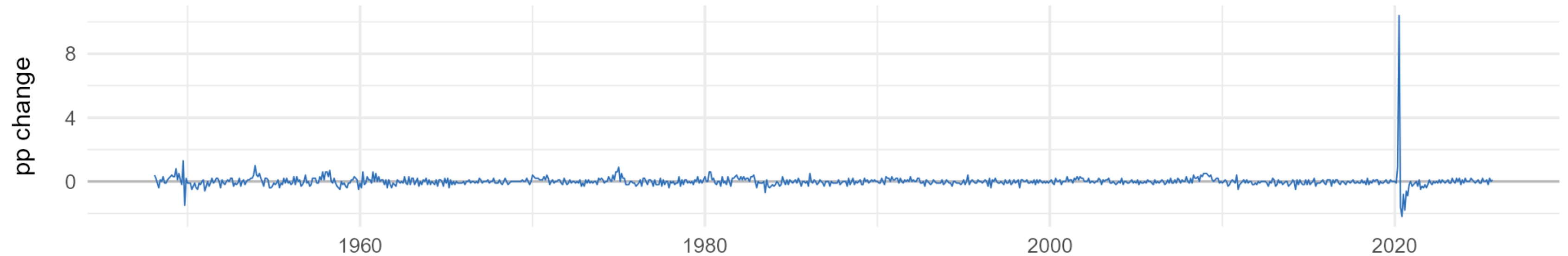
“Real evidence is never as clean as the training slides.”

ACT IV – UNRATE, CONTINUED

The series we left in Module 2

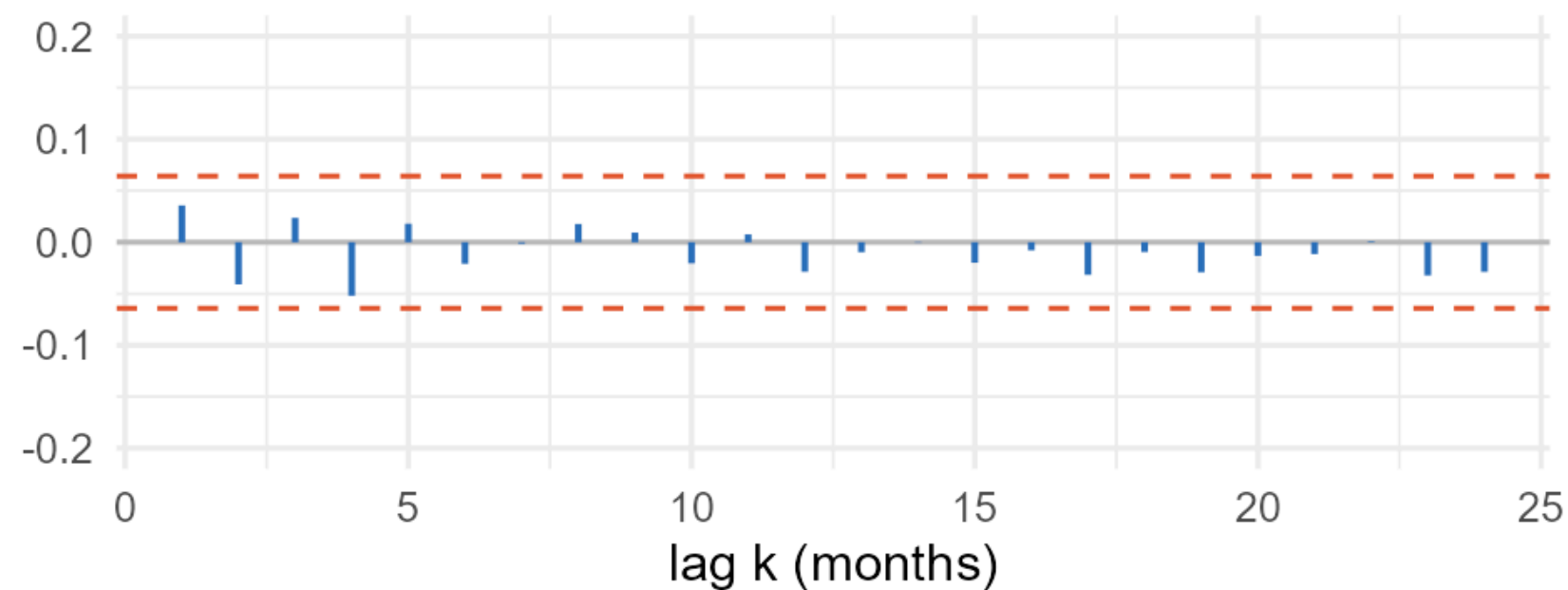
Δ US unemployment rate

monthly first difference — mean-reverting, no trend



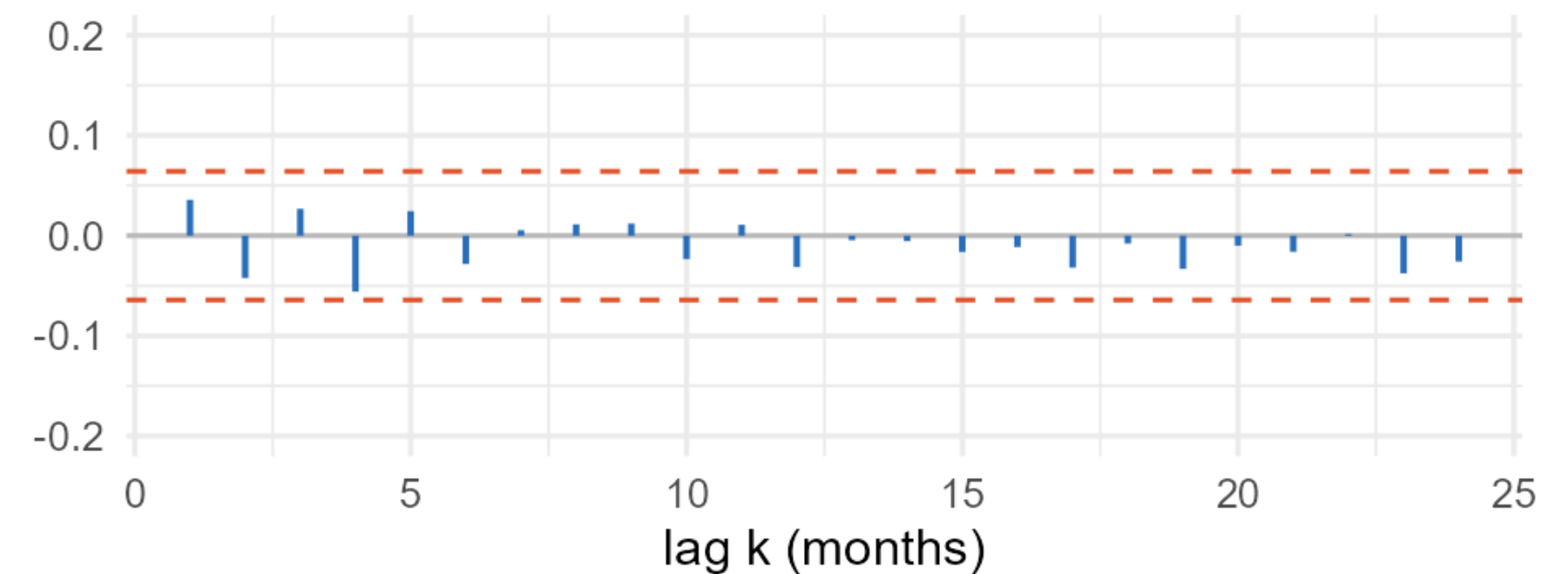
ACF

nothing outside the band



PACF

nothing outside the band



Module 2 ruled $UNRATE I(1)$, so we difference once and read the print. **Call it before I do:** AR, MA, ARMA — what order?

Nothing.

Not one autocorrelation out of twenty-four clears the band. $\hat{\rho}_1 = 0.036$, and the Bartlett band is ± 0.064 .

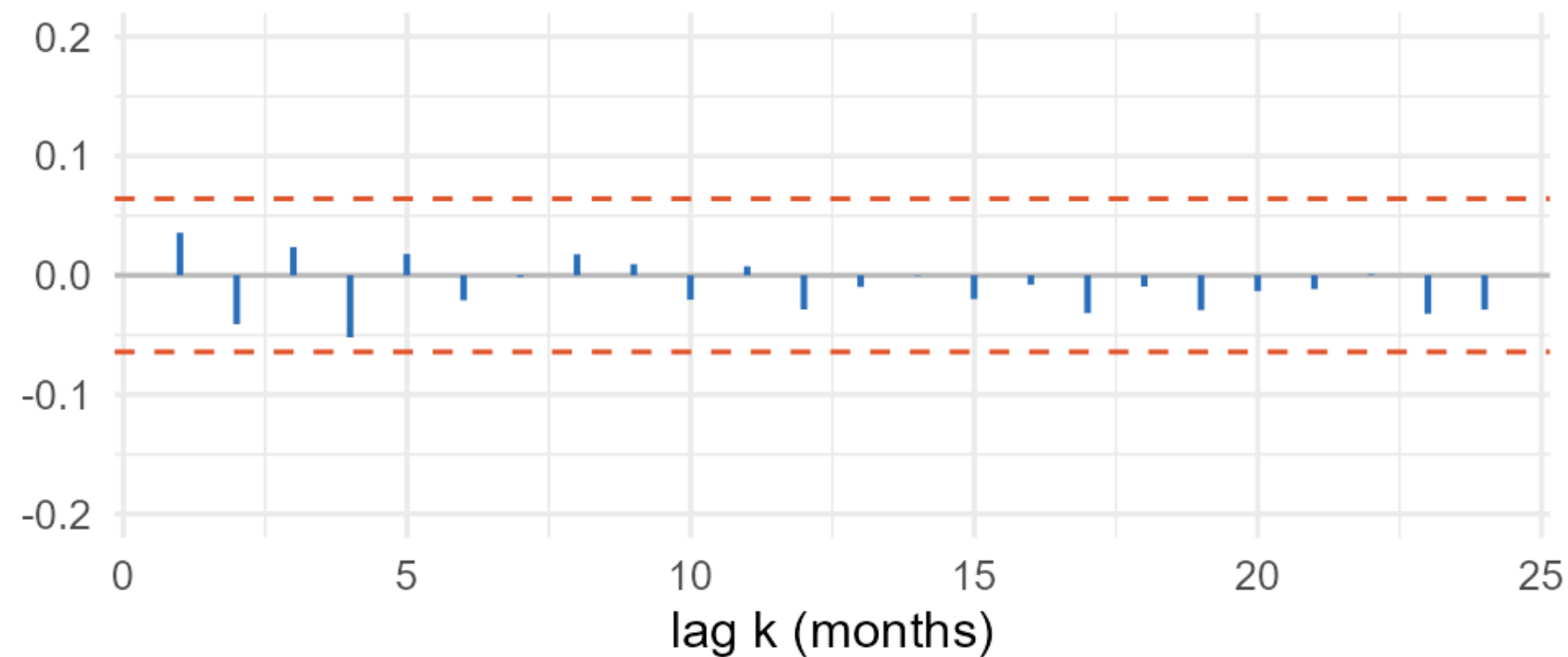
Read literally, ΔUNRATE is **white noise** — no AR, no MA, nothing to model. That should bother you. Unemployment does not turn on a coin flip.

So before believing a blank correlogram, look back at the **series**, not the bars. There is a single vertical spike in April 2020 — a **+10.4 point** move in one month.

One month ate the correlogram

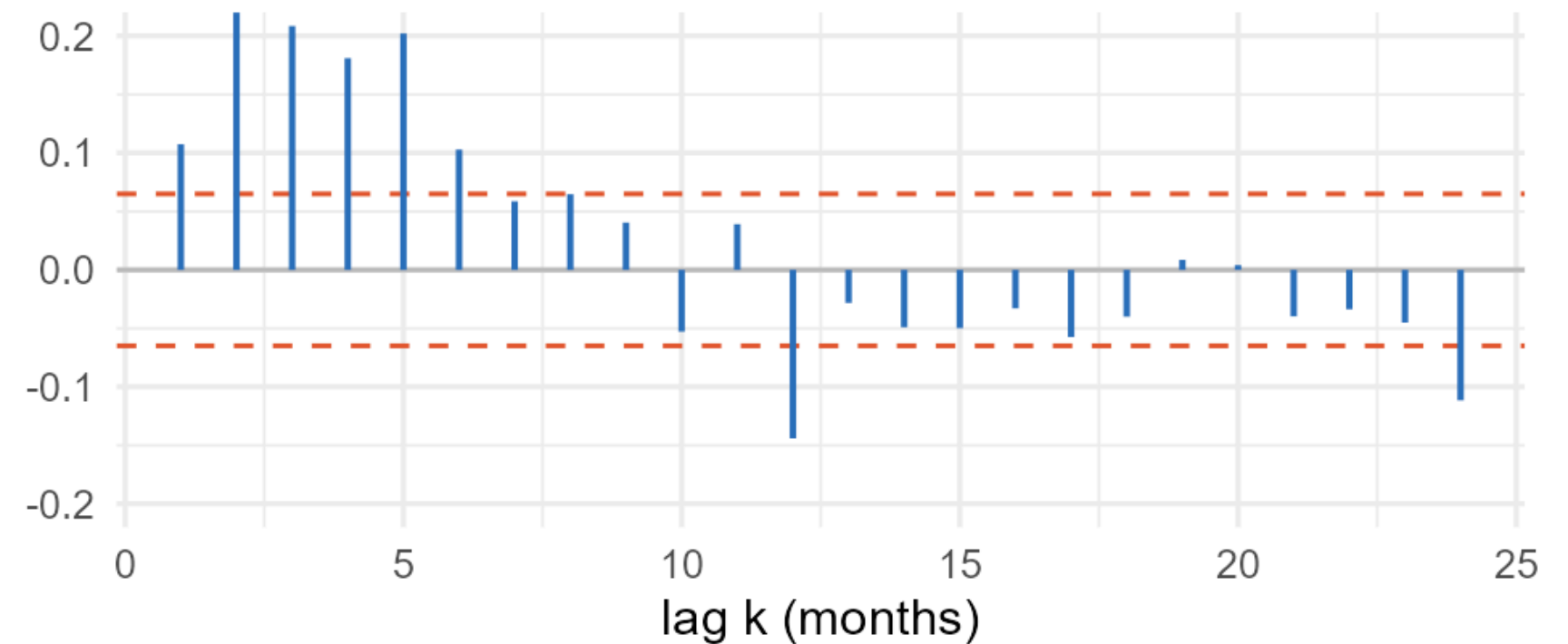
ACF — full sample

includes Apr 2020 (+10.4pp)



ACF — 2020-21 dropped

positive through lag 6, seasonal dip at 12



That one observation nearly doubles the sample standard deviation, $0.237 \rightarrow 0.415$. Every $\hat{\rho}_k$ is a covariance **over that variance**, so inflating the denominator crushes the whole correlogram toward zero.

The honest verdict

Drop the 2020–21 window and the print comes back: **8 of 24** ACF lags clear the band.

	reads
ACF	positive and significant through lag 6, peak +0.28 at lag 2 — tails off
PACF	significant through lag 5 with scattered hits out to 24, peak +0.27 at lag 2 — tails off
both	a clean negative dip at lag 12

Neither panel cuts off, so by the lineup table this is an **ARMA** — plus something annual that our table has no row for. First guess: a low-order ARMA with a seasonal term.

The table got us to a defensible guess, not to an answer.

What you can do now

1. **AR(p)** is memory of y ; stationarity needs every root of $\Phi(L)$ **outside** the unit circle. ACF tails, PACF cuts at p .
2. **MA(q)** is memory of shocks; always stationary, invertible when $\theta(L)$'s roots are outside. ACF cuts at q , PACF tails.
3. **They are the same object from two directions** — a stationary AR is an $\text{MA}(\infty)$, an invertible MA is an $\text{AR}(\infty)$. That duality is *why* the second fingerprint works.
4. **The PACF** is the OLS coefficient on the deepest lag; it counts p for you.
5. **The lineup**: tails/cuts, cuts/tails, tails/tails.
6. **A fingerprint is evidence, not a confession** — small T blurs it, one outlier can erase it, and $\hat{\theta} \rightarrow -1$ means the estimator is in trouble.

Next time

Row three of the table is unfinished business, and so is everything we just guessed at:

- **Information criteria.** When both correlograms tail off, AIC and BIC arbitrate — stop eyeballing, start scoring candidate (p, q) pairs.
- **Estimation, properly.** What `arima()` was actually doing when it pegged $\hat{\theta}$ at -1 .
- **Then Module 5** takes the lag-12 dip seriously: seasonality and SARIMA.

You can now *write down* the models. Next: how to *pick* one.